WHEAT.....

Selection of Seed :

Only healthy seeds of right variety suitable for a particular locality should be selected for sowing. There should not be any mixture of seeds of other varieties of weeds. The seed should be purchased from a reliable source. Always use certified .

If seed is not treated ,treat with vitavax, thiram @ 2.5 g/kg of seed.

varieties-

lok-1

Malavia-12

C-306

Pbw-502

K-65

Sujata

Time of sowing:

Rainfed wheat generally sown in the 2nd fortnight of October and beginning of November.

If a variety is sown later than its norm**. Long and medium duration varieties should be sown in the 1st fortnight of November and short duration al time of sowing there is adverse effect on its yield.** Under specific circumstances wheat is sown in the month of December too.

Note-

When wheat is sown in the **month of December** there is a drastic reduction in yield.

Seed rate & spacing

Usually, a seed rate of 100 kg per ha is sufficient under favourable conditions of normal sowing.

For varieties with bold grains like Sonalika and under late sown conditions, seed rate should be increased to 125 kg per ha .In case where wheat is to be sown by dibber, a seed rate of 25-30 kg per ha is sufficient.

For normal sown crop a spacing of 20-22.5 Cm between rows is recommended. When sowing is Adelayed a closer spacing of 15-18 Cm should be adopted.

Depth of sowing:-

Seeds of semi-dwarf (one gene dwarf) varieties could be sown at the depth of 5-6 cm but seeds of three dwarf varieties should not be sown deeper than 4 cm. similarly, in late sown crop, seed should be sown shallow (about 4 cm deep)as deep sowing delays the emergence of seedlings by 2-3 days and

Dibbling is the placing of seeds in holes or pits at equal predetermined distances and depths. This is done by dibble, planter or manually.

Different method of sowings :

Wheat is sown by four methods:

1. **Drilling**: In this method seed is sown by seed drill or ferti-seed drill. With the help of this implement seeds drop at uniform depth and results in uniform germination and regular stand. Seed bed should be fine and well leveled free from clods and weeds for the use of seed drill.

2. **Behind Local Plough:-**This method consist of dropping the seeds by hand into the furrows that have been opened with local plough. When seeds is dropped in furrows by hand, it is called Kera method and when it is dropped through a Pora or Nai or Hazara a special attachment with local/desi plough it is called Pora method

Seed planting in the furrow created by the plough

3. **Dibbling:** This method is used in case where supply of seed is limited. Sowing is done the help of a small implement known as Dibbler It is a wooden or Iron frame with pegs. The frame is pressed in the field and lifted and then one or two seeds are dropped by hand in each of the hole. It is not a common method because it is a very time consuming process.

Dibbling is the placing of seeds in holes or pits at equal predetermined distances and depths. This is done by dibble, planter or manually.

4. **Broadcast Method:** In this method the seeds are broadcast and then worked in by harrowing in order to cover them. However, the seeds are not uniformly distributed in the field. This method of sowing is very insufficient and should not be encouraged. Germination of broadcast seed is relatively oor and the plant stand is often irregular.

Manual Broadcasting

Fertilizer management

It was demonstrated that **120 kg nitrogen, 60 kg phosphorus and 30 kg potash per hectare** were required for optimum productivity.

The N was to be applied in two split doses of 60 kg as basal and the remaining 60 kg at first irrigation and full phosphorus and potash to be applied as basal.

Recently, the new wheat varieties have responded up to 180 kg N/ha with optima dose around 150 kg/ha. **In the Indo-Gangetic plains, application of zinc @ 25kg/ha in rice-wheat system was found to increase the yield substantially**.

Nutrient management

With intensive agriculture, deficiency of essential nutrients has also become wide spread. **At the national level, the deficiency level in micro nutrients is Zn: 46 %, B: 17 %, Mo: 12 %, Fe: 11 % and Cu: 5%.** The deficiency of sulphur has also been reported across a wide range of soils (38%).The yield response to sulphur has been obtained in more than 40 crops including cereal, millets, oilseeds and pulses etc. To realize the potential yield, strategies may include:

- Site specific nutrient management for targeted yields
- Integration of crop residues, bio fertilizers etc with inorganic fertilization
- Tillage techniques like **FIRBS** for increasing nutrient use efficiencies .

- Use of recommended doses of the fertilizers (NPK)
- Nitrogen use efficiency is more when fertilizer is placed by drill
- 1/3rd N application at sowing and 2/3rd at first node stage.
- Supplementing inorganic fertilization (NPK) with Zn and FYM increases the wheat yield.
- Application of K is necessary.

• Micronutrient deficiency appears under light soils under intensive cropping especially in rice-wheat cropping system. ⍰

In sulfur deficient soils, SSP, Cosavet -90 WDG (10 kg/ha) or gypsum (250 kg/ha) can be applied.

For increasing yield and protein content, Zinc sulphate should be applied @ 25kg/ha once in rice-wheat system or 2-3 foliar spray of 0.5% zinc sulphate (21% zinc) .

In Mn deficient soil, spray 0.5% managanese sulphate solution 2-4 days before first irrigation and two to three sprays afterwards at weekly intervals on clear sunny day.

WATER MANAGEMENT-

Application & conservation methods ,their water use efficiency ,water requirement of crop, critical stages for irrigation and probable losses if not applied : Adequate soil moisture is required for normal development of the wheat plant at all the stages of growth.

The crow root initiation stage and heading stage are crucial stages when plant suffers most due to moisture stress .

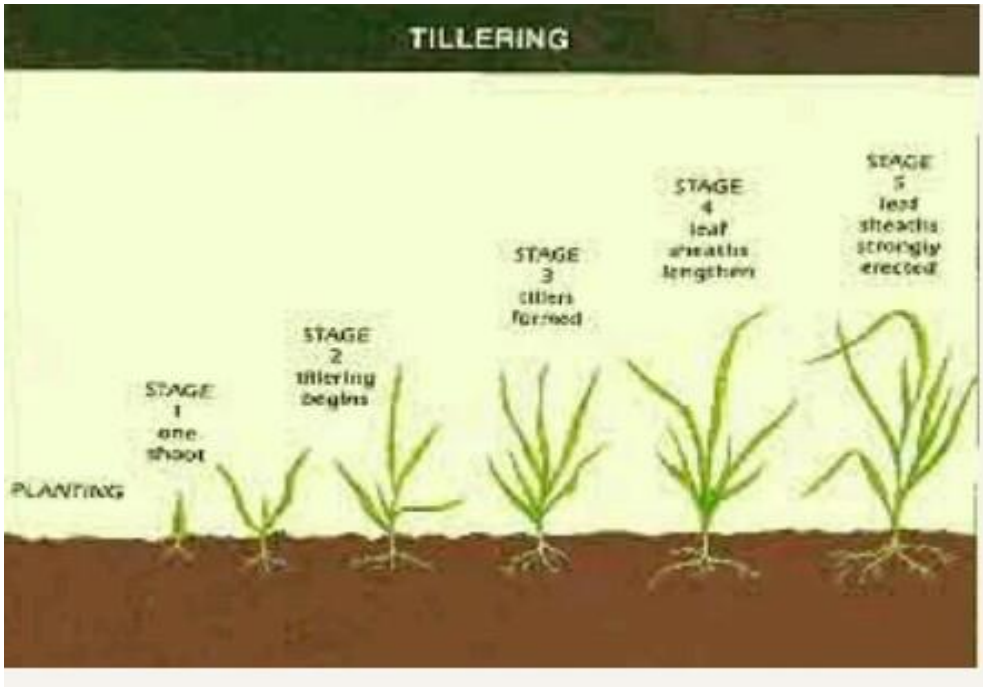
The following schedule of irrigation should be followed for dwarf varieties of wheat:

In case of dwarf high yielding varieties ,a pre-sowing irrigation should be given and crop sown when the field becomes fit for operation.

1st irrigation: The 1st irrigation to the standing crop should be given 20-25 DAS(CRI stage). In cooler regions like hilly tracts and in late sown wheat ,it is desirable to apply 1s irrigation approximately 25-30 DAS. Delay in giving this

irrigation should be avoided as it would result in upsetting the synchronous tillering in dwarf high yielding wheat varieties, abnormal heads, poor root system and finally poor grain yield. It is the most crucial stage for irrigation.

2nd Irrigation: At tillering stage, within 40-45 DAS.



3rd irrigation: At late jointing stage , within 70-75 DAS.

4th irrigation: At flowering stage ,within 90-95 DAS. Irrigation at this stage is also important because during this period plants suffer most from soil moisture deficiency . The grain number and grain size are reduced considerable.

5th irrigation: At dough stage, within 110-115 DAS.

The total number of irrigations required will vary **depending upon soil type, winter rainfall, amount of water applied per irrigation.** Under limited supply of water the following schedule of irrigation should be adopted for best utilization of available quantity of water.

Note-

1. Where only one irrigation is possible, give it **at crown root initiation DAS (CRI) stage (20-25).**

2. Where two irrigations are available, 1st irrigation should be given at CRI stage and 2nd at flowering stage.

3. Where three irrigations are a possible, 1st irrigation should be given at CRI stage and 2nd at lte jointing (boot) and 3rd at milking stage. These

recommendations strongly stress the importance of irrigation at CRI stage. It has been found that each week delay in 1st irrigation from CRI stage results in yield reduction of 200-300 kg per hectare.

Using the FIRB technology, it had been demonstrated that water saving to the extent of about 30 percent could be achieved for wheat cultivation. Source-DWR

Weed Management

Weeds emerge with the emerging crop seedlings and if not controlled in the early stages of crop growth these may cause reduction in yield varying from 10-40 % depending upon the intensity and kind of weeds present in an area.When field is generally infested from both dicot and monocot weeds .

The major dicot weeds are Chenopodium album, Fumaria purviflora,Cirsium arvense Anagalis arvensis , Melilotus alba, and Melilotus indica,Vicia sativa, , lathyrus spp.,etc., Monocot weeds include: Phalaris minor, Avena fatua, , Polypogon monspllensis, Cyperus rotundus and Cynodon dactylon.

Photo: Chenopodium album , Fat Hen. Blaxland,

(Cyprus rotandus)

Phalaris minor is the major weed of wheat in rice wheat system. Sometimes its population is so high (2000-3000 plants / m2) that farmers are forced to harvest the wheat crop as fodder.

MANAGEMENT-

2,4-D should be sprayed in wheat fields 32-35 DAS. This period may be extended to the maximum by one week.

Pre-emergence

As pre-emergence, only Stomp 30EC (Pendimethalin) is available which can be applied @ 3300 ml/ha (1000 g a.i/ha) at 0-3 days after sowing in 500 liters of water /ha. Care must be taken to have fine tilth for better performance of pendimethalin. It controls both grasses and broadleaved weeds.

Post-emergence

During the last 3-4 years a number of herbicides were found effective against even the resistant biotypes of Phalaris minor.

Out of the four new herbicides found effective against Phalaris, two namely Sulfosulfuran and Metribuzin were effective against both grassy and non-grassy weeds, whereas clodinafop and fenoxaprop were specific to grassy weeds.

Application of Metribuzin should be done carefully as this chemical is not safe at double the recommended dose. Also its application is risky if hot and windy weather prevails and rainfall occurs immediately after its spray / application.

The herbicides that are to be applied as post emergence after first irrigation at 30-35 days of sowing or 2-3 leaf stage of Phalaris minor are;

Both Grassy and Broad Leaved

1. Sulfosulfuran @ 25.0g a.i./ha in 250-300 liters of water /ha.

2. Metribuzin @ 175 g a.i./ha in at least 500 liters of water /ha.

3. A mixture of Sulfosulfuran at 25g/ha and metsulfuron methyl @ 4 g/ha in 250-300 liters water /ha.

4. Combination of 2,4-0 and isoproturon can also be used for the control of mixed weed population in resistance free area. Only Grassy Weeds 1. Clodinafop @ 400 g/ha (60g a.i./ha) in 250-300 liters of water / ha. 2. Fenoxaprop-ethyl @ 80-120g a.i./ha in 250-300 liters of water / ha.

Cultural and Preventive

Use clean wheat seed that is free from weed seeds.

Go in for early sowing of wheat (before 15 Nov.).Adopt closer row spacing (18 cm).Adopt criss-cross sowing to increase population density of the wheat plants.

Place basal dose of fertilizer 2-3 cm below the seed. Sowing of wheat on FIRBS reduces weed population.

Mechanical Control

It involves the removal of weeds by various tools & implements including hand weeding & pulling.

Diseases and their management-

Ergot

Ergot is a fungus, an ascomycete, that forms sclerotia, ascospores and conidia. The sclerotia vary in size (2-20 millimeters long), depending on the host's seed size which the sclerotia replace, and are purple-black, elongate (spur-like) with white centers. Ascospores are long, thin and septate; conidia are small, round and one-celled.

Management: Use sclerotia-free seed and implement crop rotation with non-host crops such as legumes, corn and sorghum.

Loose smut

The normal head tissue of plants infected by loose smut is completely replaced with dark masses of fungal spores, giving the heads a black powdery appearance. It is possible to see heads damaged by loose smut while much of the head is still inside the boot. Only the central stem of the head is left after the spores are released

Management: Fungicide seed treatment, disease-free seed sources.

Diseases Affecting Leaves

Bacterial streak

Early symptoms of bacterial streak include small, water soaked areas between leaf veins.

These water-soaked areas become tan streaks within a few days. When the disease is severe, streaks may merge to form large, irregular areas of dead tissue. When dew is present, the bacteria causing this disease may ooze from the lesions and dry to form a clear, thin film.

This film flakes easily and is visible when the leaf is viewed from different perspectives. Management: Avoid highly susceptible varieties .

Leaf rust

Small, orangish-brown lesions are key features of leaf rust infections. These blister-like lesions are most common on leaves but can occur on the leaf sheath, which extends from the base of the leaf blade to the stem node. Lesions caused by leaf rust are normally smaller, more round and cause less tearing of the leaf tissue than those caused by stem rust.

Management: Genetic resistance, foliar fungicides.

Powdery mildew

Powdery mildew causes white lesions on leaves and leaf sheaths. Glumes and awns also can be infected when the disease is severe. Fungal growth is largely limited to outer plant surfaces and can be easily wiped away by rubbing a finger across affected areas. Mature lesions may have dark, reproductive structures mixed with the white, cottony growth of the fungus.

Management: Genetic resistance, foliar fungicides.

Stripe rust

Stripe rust causes yellow, blister-like lesions that are arranged in stripes. The disease is most common on leaves, but head tissue also can develop symptoms when the disease is severe. Outside the United States, this disease is sometimes referred to as yellow rust.

Management: Genetic resistance, foliar fungicides.

Genetic resistance.

Wheat streak mosaic

Leaves of plants infected with wheat streak mosaic have a bright yellow streaking. Symptoms are often most severe near the leaf tip. The virus that causes wheat streak mosaic survives in volunteer wheat and is spread by wheat curl mites. The disease is often most severe in areas of a field that are closest to these sources of the disease and mites. Commonly, plants infected with wheat streak mosaic are also infected with High Plains disease and Triticum mosaic. The symptoms of these diseases are nearly identical. Disease severity is greater when plants are infected by more than one virus.

Management: Control volunteer wheat, delayed planting date, genetic resistance, avoid planting near maturing corn.

Insect and Management

Wheat Aphid

Aphid Aphids are small, soft-bodied insects that obtain their nutrition by sucking sap from plants. They have two cornicles that protrude from the upper surface of the abdomen near the tail. These “tailpipes” vary in length and may be reduced to mere bumps in some species. During feeding, aphids inject salivary secretions into plant tissues. The salivary secretions of some aphids are toxic and can cause severe tissue damage in the plant.

Symptoms of aphid

injury include leaf stippling, discoloration or striping. In addition, wilting, premature browning and death of the plants may result. Management: Controlling volunteer wheat, avoiding early planting of winter wheat .

Management-

Monitoring aphid populations in the fall and spring and following treatment guidelines are necessary to minimize the risk of serious losses.

Wireworms

These insects may spend the winter as adults or larvae. Adults are about 1/4 to 3/4 inch long, hard shelled, brown to nearly black with a streamlined body. The adults are called click beetles because when they are on their backs, they will flip up and turn over with a clicking noise. Larvae live from two to six years in the soil feeding primarily on roots. Wireworms eat the germ of the seeds or hollow them out completely, leaving only the seed coat.

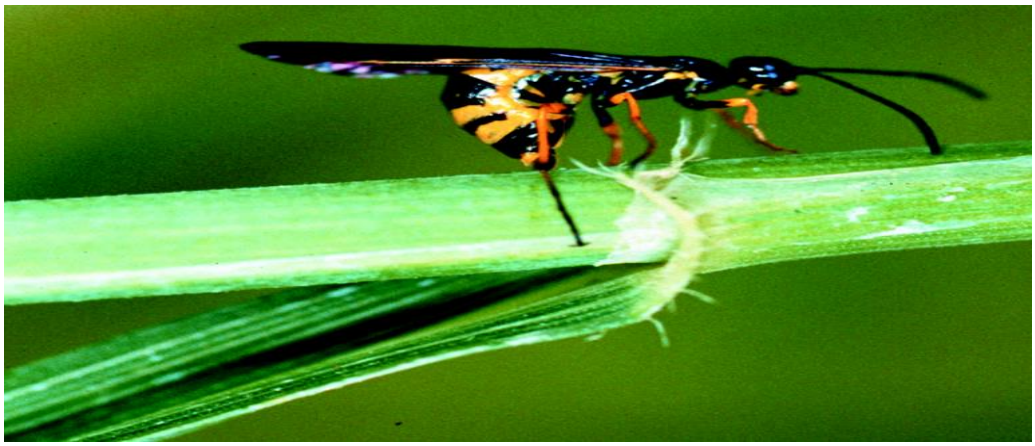
Adult wireworms, known as "click beetles,"

Wireworm Larvae

Management: Wireworms are among the most difficult insects to control. Planter box insecticide seed treatments will reduce damage to the seeds, but will not protect seedlings.

Sawfly

The adult sawfly is a slender wasp about 1/2 inch long with a black abdomen and yellow ringed markings.



When observed in the field, sawflies may be recognized by the unusual habit of resting head downward on the stems of grain plants.

Sawfly damage is threefold. First, they cause grain yield reduction by their tunneling activity in the infested stems.

Additional loss occurs when sawfly-cut stems fall to the ground and become unharvestable. The protein content of grain from the infested stems is also lower. Management: Crop rotation, delayed planting and fall tillage.

Armyworms

In autumn, fall armyworms may be a problem in seedling plants. Fall armyworms feed on the young plants and eat the plants to the ground, causing a loss of stand. armyworms will move up from leaf feeding and cut the wheat heads from the plant stem.

Management: An infestation level of five to six worms per square foot will justify treatment. Treat only with liquid insecticides during the warmer part of the day.

Harvesting , threshing & storage:

harvesting and threshing methods (manual/mechanical), storage-moisture content , drying, chemical used for storage pests.

Harvesting and threshing:

High yielding dwarf varieties of wheat should be harvested when the leaves and stems turn yellow and become fairly dry.

The right stage for harvesting is when there **is about 25-30% moisture in grains**. Harvesting is normally done with serrate edge sickles by hand. Bullock driven reapers are also used occasionally .

Combines are also available which can be harvesting, threshing and winnowing wheat crop in single operation .

After harvesting the crop by hand, it is dried and three to four days on the threshing floor and then threshing is done by trampling bullocks or thresher attached to bullocks.

Now-a-days power driven stationary threshers are becoming more popular because these are easy in operation and hasten the process.

Yield: When cultivation of high yielding dwarf varieties of wheat is done with improved scientific methods, they produce about 45- 55 qtl of grain per hectare. under irrigated conditions.

20-25 qtl per hectare under rainfed conditions.

Crop Products

i. Industrial use of product and by-product, Products

Whole wheat flour

It contains the finely ground bran, germ and endosperm of the whole kernel. Whole wheat products have a distinctive flavour and coarser texture than those made from white flour.

Maida:

Wheat bran:

By-Products

We would like to consider using wheat by-products, mainly stalks, for the following products; (1) geotextiles, (2) filters, (3) sorbents, (4) structural composites, (5) non-structural composites, (6) molded products, (7) packaging, and (8) combinations with other materials.